

## Exercise 7: Financial Forecasting

```
public class FinancialForecasting {  
  
    public static double Value(double currentValue, double growthRate, int years) {  
  
        if (years == 0) {  
            return currentValue;  
        }  
  
        return Value(currentValue, growthRate, years - 1) * (1 + growthRate);  
    }  
  
    public static void main(String[] args) {  
  
        double presentValue = 10000;  
  
        double growthRate = 0.10;  
  
        int years = 5;  
  
  
        double futureValue = Value(presentValue, growthRate, years);  
  
        System.out.println("Future Value after " + years + " years: " + futureValue);  
    }  
}
```

The screenshot shows an IDE with the following components:

- Editor:** Displays the `FinancialForecasting.java` file with the following code:

```
1 public class FinancialForecasting {  
2     public static double Value(double currentValue, double growthRate, int years) {  
3  
4         if (years == 0) {  
5             return currentValue;  
6         }  
7  
8         return Value(currentValue, growthRate, years - 1) * (1 + growthRate);  
9     }  
10  
11     public static void main(String[] args) {  
12  
13         double presentValue = 10000;  
14  
15         double growthRate = 0.10;  
16  
17         int years = 5;  
18  
19  
20         double futureValue = Value(presentValue, growthRate, years);  
21  
22         System.out.println("Future Value after " + years + " years: " + futureValue);  
23     }  
24 }  
25
```
- EXPLORER:** Shows the project structure with folders like `EngineeringConcepts` and `DesignPatterns`, and files like `FinancialForecasting.java`.
- TERMINAL:** Shows the execution of the program:

```
Tharun@LAPTOP-1791M027 MINGW64 /f/Coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/Algorithms_DataStructures (main)  
$ javac FinancialForecasting.java  
Tharun@LAPTOP-1791M027 MINGW64 /f/Coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/Algorithms_DataStructures (main)  
$ java FinancialForecasting  
Future Value after 5 years: 16105.100000000008  
Tharun@LAPTOP-1791M027 MINGW64 /f/Coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/Algorithms_DataStructures (main)  
$
```